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Artificial intelligence for education and its emphasis on assessment and adversity quotient: a review

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Abstract

Purpose – The purpose of this study was to analyze importance of artificial intelligence (AI) in education and its emphasis on assessment and adversity quotient (AQ).

Design/methodology/approach – The study utilizes a systematic literature review of over 141 journal papers and psychometric tests to evaluate AQ. Thematic analysis of quantitative and qualitative studies explores domains of AI in education.

Findings – Results suggest that assessing the AQ of students with the help of AI techniques is necessary. Education is a vital tool to develop and improve natural intelligence, and this survey presents the discourse use of AI techniques and behavioral strategies in the education sector of the recent era. The study proposes a conceptual framework of AQ with the help of assessment style for higher education undergraduates.

Originality/value – Research on AQ evaluation in the Indian context is still emerging, presenting a potential avenue for future research. Investigating the relationship between AQ and academic performance among Indian students is a crucial area of research. This can provide insights into the role of AQ in academic motivation, persistence and success in different academic disciplines and levels of education. AQ evaluation offers valuable insights into how individuals deal with and overcome challenges. The findings of this study have implications for higher education institutions to prepare for future challenges and better equip students with necessary skills for success. The papers reviewed related to AI for education opens research opportunities in the field of psychometrics, educational assessment and the evaluation of AQ.

Keywords Education, Artificial intelligence, Teaching-learning analytics, Assessment, Adversity Quotient

Paper type Literature review

1. Introduction

Modern technology is conquering "artificial intelligence" (AI) (Chen *et al.*, 2020a), which is the center of significant computer research, design and application. Humans are trying hard to embed AI in a machine by making use of natural intelligence. Why? Because it has been proved by earlier researchers, the need for time competes with smart systems (Carbonell, 1970; Sarker, 2022; Spector and Ma, 2019), which take less effort to operate, work automatically, efficiently, etc. The field is in demand in the 21st century characterized by many varied and independent efforts. Intelligence (Atrizka and Pratama, 2022) means the capability to attain and apply information, knowledge and skills. AI does not exactly work like human cognitive intelligence, and it is closer to the realm on which this research review focuses. AI is spanning the education sector speedily. Education is multidimensional (Ouyang and Jiao, 2021). Education accommodates administration, academic monitoring, counseling, teaching-learning, assessments, exams and feedback. Some policies help to improve quality of education. Recently Indian education System devised New Education Policy in 2020 (NEP) (Aktar, 2021). It is introduced as NEP 2020. It signifies skill-based assessment. NEP 2020 talks about the 360-degree development of a student which is incorporated in the following section. NEP focuses on talent, capabilities and life skills. This literature discusses about AI-based assessment framework to evaluate the adversity quotient

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(AQ) of a student with the help of technology. Table 1 consists of abbreviations used in the paper to understand the term's meaning.

A. Motivation

I. In Goal 4 [Sustainable Development Goal (SDG) 4] of the 2030 Agenda (Zhang *et al.*, 2020; Cf, O. D. D. S., 2015), the worldwide education expansion agenda is reflected for sustainable development, embraced by India in 2015 – in pursuit of "guaranteeing all-encompassing and impartial quality education and promote learning opportunities for all which are lifelong" by the year 2030 (Vinuesa *et al.*, 2020). In effect, all of the critical goals and objectives (SDGs) for sustainable development could be accomplished by 2030 (Morrow-Gorton, 2022).

It is an urge of qualitative, multidimensional, multidisciplinary education, which must focus on modernizing, acclimating and fascinating new significant viewpoints in unique and varying arenas.

II. Education: Education is the most approachable aspect used to develop and improve natural intelligence. Many scientists (Pathak, 2007) have interpreted the term *Education* from their perspective.

Table 2 gives different definitions that articulate the meaning of "Education". After proceeding with the summarized meaning of these definitions, it can be outlined as "Education is a continuous process of learning with the aim of cultivating knowledge and imbibe life-skills to become an ideal citizen."

III. Role of National Education Policy [NEP] 2020 in current education:

National Education Policy (2020), Aktar (2021) is the primary education strategic program of the current century. Its objective is to address the many growing developmental certainties of India in the first and foremost place. This policy recommends the amendment and renovating

Abbreviation	Meaning
NEP 2020	National/New Education Policy 2020 (India)
SDG	Sustainable development goal
AI	Artificial intelligence
NLP	Natural language processing
NN	Neural network
ML	Machine learning
DL	Deep learning
GA	Genetic algorithm
ASM	Advanced statistical model
IEEE	Institute of Electrical and Electronics Engineers
DARPA	Defense Advanced Research Projects Agency
AQ	Adversity quotient
MPSC	Maharashtra Public Service Commission
ICAI	Intelligent Computer-Aided Instruction
GALA	Genetic Algorithm Learning Application
RACE	Reading comprehension dataset from examinations
BLEU	Bilingual evaluation understudy
GRE	Graduate record examinations
CET	Common entrance test
AIED	Artificial intelligence in education
HE	Higher education

Table 1.
List of abbreviations

Sr. No.	Scientist	Education
1	Aristotle (384–322 BC)	"Education is the method of training a person to fulfill his intention by exercising all the abilities to the fullest extent as a member of society."
2	Martinus J. Langeveld (1905–1989)	"Education is every interaction that happens is every association that occurs between adults with children is a field or a state where the educational work in progress (Standish and Saito, 2012)"
3	Prof. H. Mahmud Yunus (1899–1982)	"In order for the child to live a happy and all what he did be beneficial to himself and society (Srimulyani, 2011)"
4	Socrates ((469–399 BC)	"Education means bringing out the ideas of universal validity which are latent in the mind of every man"
5	Mahatma Gandhi (1869–1948)	"By education I mean an all-round drawing out of the best in child and man-body, mind and spirit"

Table 2. Various definitions of "Education"

of all characteristics of the education structure as shown in Table 3. NEP 2020 is based on the trust that education must develop:

(1) Cognitive abilities

(2) Communal, moral and emotional (expressive) capacities and temperaments.

This structure consists of guidelines and governance, to create a novel system that is affiliated with the aspirational goals of existing education, including SDG4.

A. Objectives

The fundamental research objectives:

- To elaborate AI techniques used in the education sector.
- To study the significance of the assessment strategies in teaching-learning.
- To establish the relationship between assessment and AQ with the help of AI.

B. Prior Research

1. In education, intelligent quotient (IQ) has been considered an important factor for success in life. There is a new quotient called AQ which is an indicator to measure

National education policy *year 1968*	National educational policy *1986* (Revised in the *year* 1992)	National educational policy year *2020*
1. *The "Education to all"* vision 2. Focused on differently abled students and *Girl's* education	1. This policy is motivated by *Early Childhood Care* and *Education* (ECCE) through a *child-centered approach* (CCA) 2. Setting up *autonomous colleges* and *countryside universities* in rural India followed by Gandhian ideology	1. It is designed to enable complete essential learning 2. *Universal right of entry to education* which leads towards the 100% Gross Enrollment Ratio (GER) 3. *Fundamental principles are* 3.1. Distinguish, categorize and foster the exceptional capabilities of each student 3.2 Life skills- communication skills, collaboration skills, cooperative in a team, supportive skills and flexibility

Table 3. National educational policies analysis (India)

resilience. There is a connection between AQ and problem-solving skills (Oliveros, 2014), resilience (Matore *et al.*, 2021), critical thinking (Spector and Ma, 2019) and adaptive behavior (Tomaszewski *et al.*, 2018). With the help of assessments, life skills can be measured. With the help of AI techniques, it is possible to evaluate the AQ (Naimnule *et al.*, 2020; Wang *et al.*, 2021) of a student. The purpose of this research was to exemplify a wide-ranging vital effort of AI techniques used to perform a specific task in the education segment.

AI techniques, assessment patterns and adversity skills deliberated and summarized over twenty multidisciplinary published available research work from 2013 to 2023. The study provides a literature review on techniques for systematically teaching-learning analytics. The effort aims in categorizing several methodologies, wide applications and assessment processes as listed in Table 4.

C. Research Goal

2. Study emphases on multiple domains such as education, the role of AI in education (Chen *et al.*, 2020a), the Significance of assessment (Halachev, 2013; Jin, 2022; Luckin *et al.*, 2017; Zhu *et al.*, 2020), the relationship of AQ with IQ (Widodo *et al.*, 2022), EQ and SQ (Singh *et al.*, 2021; Pong and Lam, 2023) and how the AI is useful to evaluate the AQ of a student. The proposed research questions are quantified in Table 5.

D. Contribution of work

Compiled study of 141 bibliography articles were engrossed in classifying evidence which was required to answer mentioned research questions *vis-à-vis* the six methodological components. These components of AI constitute adopted deep learning (DL), machine learning (ML), neural network (NN), natural language processing (NLP), genetics algorithm (GA) and Fuzzy techniques. Assessment types practiced in higher education and the facet of academic assessment in teaching learning; and AQ and its role in frequent educational activities. It yields the contemporary practice and findings from the 141 papers and accomplishes the analysis through numeral references for AI technique development, for evaluating AQ. Literary surveys highlight important areas for supplementary research. This survey covers current AI trends, application-oriented techniques and the classification of AI methodologies which shows a proper knowledge-based framework for evaluating AQ with suitable techniques.

E. Organization of paper

This paper describes the significance of AI in teaching learning and the categorization of techniques. Different algorithms are developed based on applications, input data, expected output, etc. This paper emphasizes the general AI framework used in teaching-learning and assessment techniques developed in detail. Subsequently, it is a significant undertaking to assess the student to evaluate the AQ. It will help the researcher to decide on appropriate assessment technique and metric used for evaluating the performance of the student. Figure 1 reveals the paper systematized into miscellaneous sections.

2. Review strategy

A. Materials and Methods

The "*education*" word is common all over the world, the educational policies are different countrywide. Leading from the NEP draft, it is well understood that AI is the imminent key player in education. The documents were carefully chosen for the investigation of AI techniques which cater to Journals, research articles and conference papers of publication

Paper	Applied methodology	Limitations	Inferences for the future upgradation
Indian NEP Final draft (2021)	The focus is given on consistent *formative assessment* for educational learning that inspires today's 'coaching culture'	Faculty will *not have the independence to modernize* the curriculum and assessments within a narrow framework of higher education	Accredited bodies and agencies will design and *implement assessment frameworks* encompassing the design of rubrics, proficiencies, portfolios, standardized assessments and assessment analytics
Yan *et al.* (2020). Including learning analytics in the loop of self-paced online course learning design. *International Journal of Artificial Intelligence in Education*, 31, 878–895	Learning Analytics (LA) conveys immense potential in monitoring, prediction, tutoring, comment-based *feedback, adaptation, personalization*, approval and reflection	The *lack* of LA consideration in designing the current course can bring challenges to LA implementation due to the *data collection and pedagogical connection* requirements	The questions in the *formative assessment required to be auto-graded for appraising diverse cognitive skills*, so that the high-tech adaptive testing instrument can be exploited
Pelánek (2022). Adaptive, intelligent and personalized: Navigating the terminological maze behind educational technology. *International Journal of Artificial Intelligence in Education,* 32(1), 151–173	Characteristics of Edtech that are concerned with personalization, *adaptation*, or other categories of intelligent behavior	Some *student-specific personal data and state estimates* will be naturally used by a learning system to achieve personalization	Learning *Models also predict future performance frequently* and these predictions are used to evaluate and compare analysis studies
Mehraj and Baba (2019). Scrutinizing artificial intelligence-based career guidance and counseling systems: an appraisal. *International Journal of Interdisciplinary Research and Innovations*, 7(1), 402–411	Artificial Neural Network (ANN) epitomizes that in terms of *error tolerance, parallel function,* flexibility and stability	A greater number of individuals are *opposed to familiarizing themselves with upcoming industries and occupations* that promise new career prospects in addition to employment opportunities	Solutions established on *deep learning* can be developed in such expert systems to make them smarter and more efficient. *Appropriate data should be selected and analyzed* which can construct a customized standard data set
Holmes *et al.* (2019). Artificial intelligence in education. Globethics Publications	There is a major impact of artificial intelligence on *assessment,* contributing some tempting glimpses into the future	Rapid advances in AI products and services will outperform education systems' is a *real risk.* Appropriate measurements to *understand, achieve, manage and integrate* the products and services are required	Computer researchers, educational scientists, AI engineers and big tech companies have to decide *how artificial intelligence unfolds the petals in education*, or they can engage themselves in productive dialog

(*continued*)

Table 4. Survey of the education system and its future inferences from diverse author's outlook

Paper	Applied methodology	Limitations	Inferences for the future upgradation
Halachev, P. NEURAL NETWORK MODEL FOR ASSESSMENT COMPUTER-BASED TEST	The recommended *algorithm* permits a more accurate assessment of students' knowledge	To *increase the objectivity* of the study, numerous values of these criteria were required to be *synchronized between teachers* teaching identical subjects	ANNs are largely used in innumerable arenas (robotics, research, automobile, manufacturing, etc.). But *still have limited and restricted applications in the education sector*
Vincent-Lancrin and Van der Vlies (2020). Trustworthy artificial intelligence (AI) in education: promises and challenges. 2020	Artificial Intelligence-based solutions *try to assess* how individuals think and answer to a learning situation. AI tries to *adapt* to the student's needs and skills	Presently, the deployment of AI in education remains *incomplete at the system level*. Employed AI could lead to involuntary decisions or suggestions which are expected	It is challenging to prepare teachers and learners as *revolutionary agents of work and society, particularly by developing their multilayered thinking skills such as original or critical thinking*
Chen *et al.* (2020a). Artificial intelligence in education: A review. Ieee Access, 8, 75,264–75278	The study seeks to judge by what means *AI has affected teaching-learning*, governance, tracking and management areas of education	Learning analytics prepares us to *transfer from our comfort zone towards a broader scope* including inter- and intra-personal skills, arts and literature	AI Systems thrive on *all features of students*, i.e. personal skills, command of knowledge and expertise, learning aptitude and professional expansion
Tomaszewski *et al.* (2018). Adaptive behavior, executive function and employment in adults with Down syndrome. *Journal of Intellectual Disability Research,* 62(1), 41–52	The distinct pattern of *relative strengths and trials in adaptive actions* and Executive Function (EF)	A s*mall sample* of primarily young adults. Individuals establish *complications with* aspects of executive function (EF) and *adaptive behavior* across the life expectancy	Precise facets of the Down Syndrome cognitive outline may have an imperative influence on employment positions in mainly young adults

Table 4.

year 2013–2023. The co-occurrence analysis of keywords used for reviewing the papers using VOS Viewer (Wong, 2018) is shown in Figure 2. Keywords existing more than 5 times are considered for the primary analysis One of the important dimensions of education is teaching-learning. Teaching learning constitutes assessment and assessment should be to quantify the AQ, which represents lifelong learning skills.

The AI does not depict the teacher's role is compromised; it articulates the manual work done so far by the teacher has to be reduced by technology. It is a suitable to evaluate the impact of AI on contributing factors of education that are popular in teaching-learning.

B. Search Strategy: Inclusive- Search continued with Keywords (AI, education, teaching-learning) on different platforms comprising IEEE, ScienceDirect, Scopus, Web of Science, popular journals in the education domain, Google Scholar till 2023 and the number of survey papers referred results are shown in Figures 3 and 4 below.

C. Search Strategy: Exclusive-

The journals which have an H-Index of less than 10 are excluded from the study. A higher H-Index indicates more reputable journals and well-known authors worldwide. After an

Sr. No.	Research question (RQ)	Discussion
RQ1	How does artificial intelligence benefit the education domain?	It is essential to study AI perspectives (what, why and how) in education. Each AI technique is elaborated to determine its significance specific to teaching-learning
RQ2	What are the multimodal assessments conducted in higher education?	For effective assessments, it is essential to examine and explore the type of assessment, parameters used and objective of the assessment
RQ3	What are the diverse measurement metrics to affect adversity in education?	A review was conducted systematically with a metric characteristic and context in the current education system
RQ4	What is the relevance of the Adversity Quotient in the Indian context and Western contexts?	Country-wide surveys and articles have been studied to conclude similarities and differences
RQ5	Mapping assessments with adversity quotient would be a good step in higher education?	There is a substantial relationship between academic performance and adversity quotient. Academic performance and skills are measured through various assessments
RQ6	Which is the various artificial intelligence-based approaches to evaluating adversity quotient?	The quotient is defined as a degree or amount of a specified characteristic. It is a measurement of intelligence. It is possible to measure natural intelligence with the help of artificial intelligence

Table 5.
Literature based on Research Questions

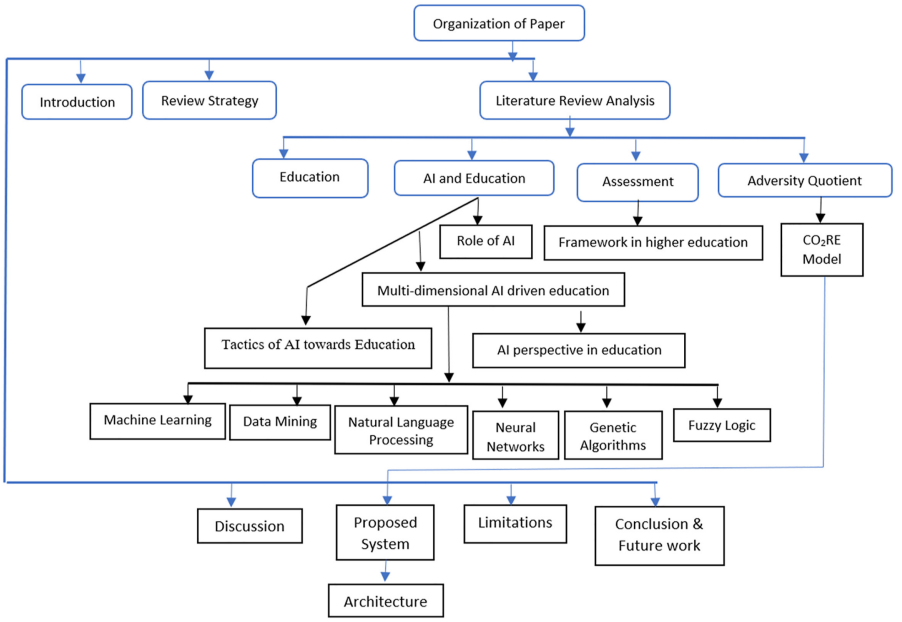


Figure 1.
Systematic structure of literature review

elimination process, more than eighty reviewed journals, research articles, books, blogs, professional publications and government and institutional reports were selected.

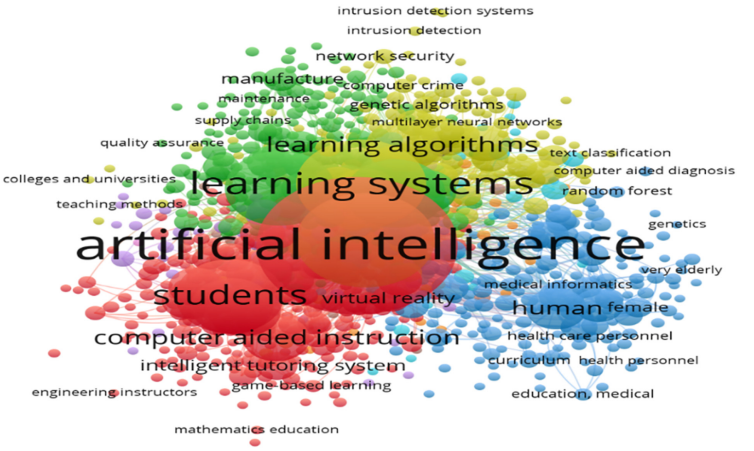


Figure 2.
Keyword co-occurrence: artificial intelligence, education and learning

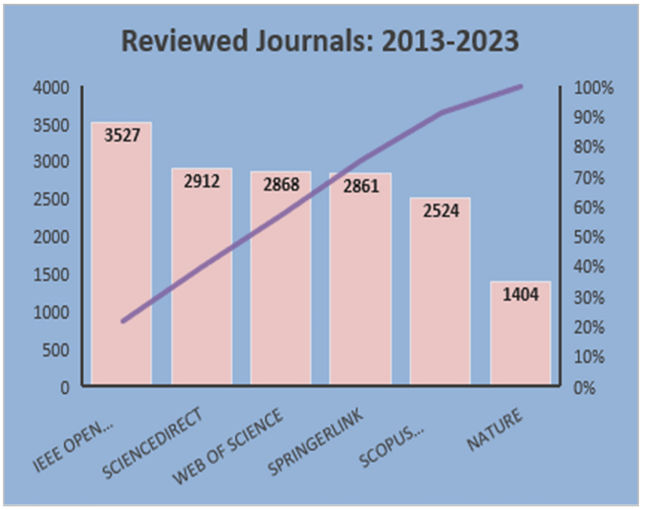


Figure 3.
Publication-wide reviewed journals

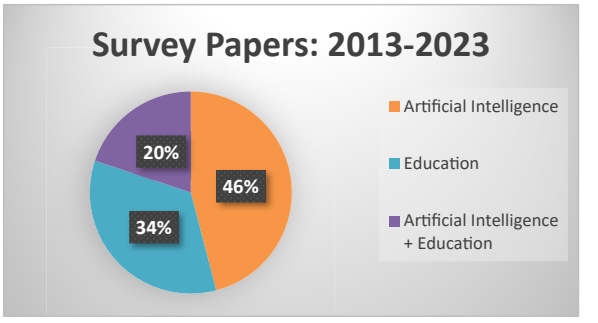


Figure 4.
Investigation of survey papers

D. Cross-section: Teaching-learning criteria

Teaching learning came along a long way and AI came into the picture in 1956 (John McCarthy) that's why, reviewed documents with the published years 2013–2023 are preferred on an aforesaid criterion. It is observed that AI made successful imprint on education, composed with teaching-learning, narrowing down the number of articles for analysis to eighty. A sample size considered for the literature study is sufficient to inform conclusions and inferences about the impact of AI techniques in teaching-learning, captivating an open approach. Figure 5 presents cross-section statistics of AI concerning education and teaching-learning analytics. A further analysis-oriented review focused on all cognitive quotients. Figure 6 classify articles from the popular journals and concludes there is too less research carried out with respect to AQ.

Figure 7 depicts AQ infusing an AI to analyze assessment data and mine information from it.

3. Systematic literature review: AI in higher education

A. History of AI: AI was invented by American computer scientist John McCarthy in 1956.

The primary employment of AI techniques in computer-aided instruction is frequently attributed to Jaime Carbonell, who introduced a system called SCHOLAR in his 1970 Ph.D. thesis (Carbonell, 1970). It is the first Intelligent Tutoring System. In this system, AI models are highly simplified computational representations (in semantic networks, as interpreted by SCHOLAR, by using ontologies or knowledge graphs).

B. Role of AI

AI is appropriately admired in accounting and finance, the stock market, agriculture, healthcare, supply chain management and education. Teaching-learning is the most crucial process to handle. Teaching includes the creation of course material, teaching, attendance, defaulter list, notifying the parents about the wards' attendance, creating summative and formative tests, conducting tests, displaying results, at last marks entry in the system for

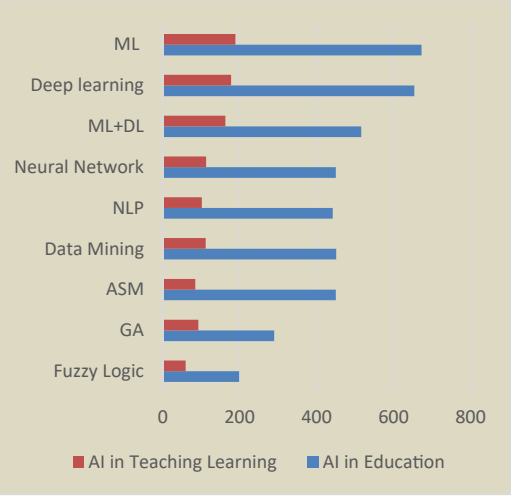


Figure 5.
Statistics of AI techniques for teaching learning and education

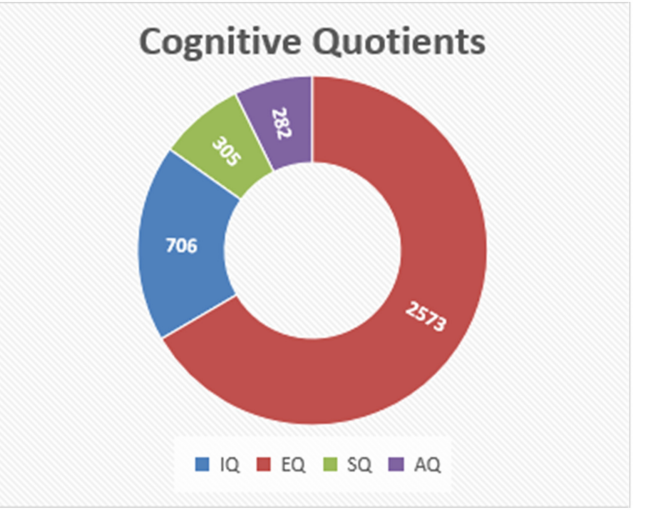


Figure 6.
Research scrutiny: Iq EQ SQ and AQ

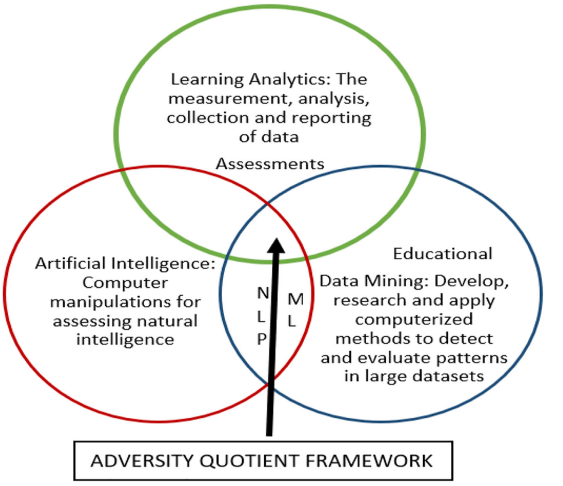


Figure 7.
Intersection of AQ related recent research areas

result generation. While learning, the student should get course material on time, continuous assessment tests, teacher feedback, subject feedback, survey, etc. If you think of a single step in a teaching-learning process, data is the fundamental aspect. Gathered data can be structured or unstructured.

Looking at the flow of Figure 8, the journey from data gathering to wisdom is possible through intelligence and embedded intelligence in computing machines could be depicted by AI.

C. Multidimensional AI-driven Higher Education

New notions for AI collection, processing, storing, analyzing, handling, visualizing and, most importantly, conventional practices, are emerging with data-driven business models currently being created around them.

How is it possible without the help of technology?

1. Extensive use of technology is used in the teaching-learning process. AI-based software has to be established (Ouyang and Jiao, 2021) and used by teachers and students.
2. Technology-based system will guide students with valuable information and supports them in making optimal career choices. But the question arises what is AI and how does it work (Chassignol *et al.*, 2018)?
3. AI (one of the domains of Computer Science) refers to the thinking style of human intelligence that is programmed in machines to ponder like humans and mimic their actions. AI is a broad domain that can be used to assess knowledge, skills and attitudes.

AI-based software to track the progress of students to empower various teaching-learning stages is shown in Figure 9. It's one of the technological challenges to integrate "AI" with Education. Though it seems to be easy, it is quite difficult to measure human intelligence with the help of AI.

To strengthen the idea, numerous national as well as international papers have been surveyed to represent the challenges towards technology as well as education point of view.

D. Tactics of AI towards Education:

If a question strikes "What is the secret of animal intelligence?" People are trying hard to identify the answer to this question but no one has succeeded in finding the "secret!" How to

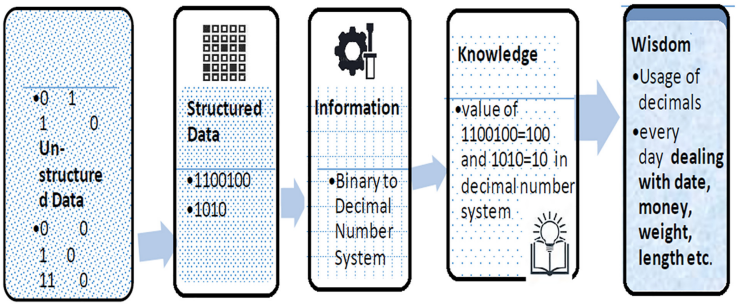


Figure 8.
Journey from data to wisdom

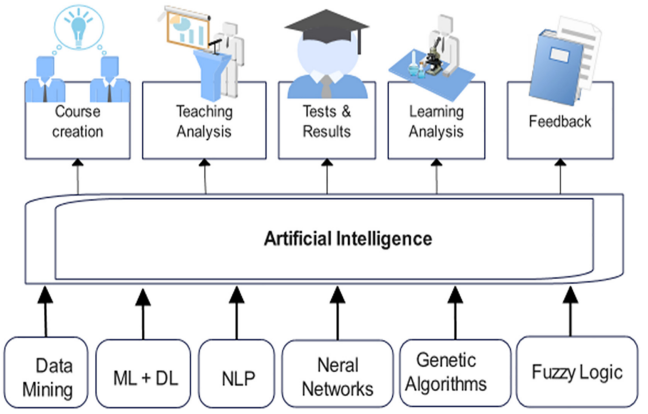


Figure 9.
AI-empowered education

learn about intelligence from animals is to study their behavior, predominantly problem-solving behavior.

Problem-solving agendas claim heavy training in puzzles and games to refine the techniques. Table 3 discusses education policies which are facing challenges and needed to apply on problem-solving strategies. If this study solves some of the problems faced by today's education system (Srimani and Patil, 2014), definitely the study is contributing not a flower but a petal towards society.

E. AI Perspective in the field of education:

There are three concerns in education that need to focus on:

1. What is the importance of AI in education?
2. Why does it make more sense to use AI for education?
3. How AI would help to support education?
    1. What:
        1.1 Chatbots Based Learning (Mondal *et al.*, 2018).
        1.2 Recommendation System for Personalized Learning (Pelánek, 2022).
        1.3 E-Learning and Distance learning (Sharma *et al.*, 2019).
        1.4 Adaptive learning and Assessments depend on the learner's level of understanding.

    There are 3 types of learners: fast, medium and slow. According to the learner, the pacing system has to adapt the assessment strategy (Jin, 2022; Luckin *et al.*, 2017).

        1.5 Exam Invigilation (Thampan and Arumugam, 2022).
        1.6 Exam Conduction in a fair way to avoid copy cases.
        1.7 Feedback through an expert system for improvement and progress (Zhu *et al.*, 2020).

    2. Why:
        2.1 AI is convenient to use and adapt: It is the era of data and technology. Everywhere there is apps usage on fingertips or mouse-clicks. e.g. Alexa, Siri, Global Positioning System (GPS) tracking system and translators.
        2.2 It reduces manual work: Not only AI, any technology helps to lessen the burden on humans.
        2.3 Err-free: Humans can make mistakes sometimes but a trained model of AI is error-free.

It has been quite convincing what and why. In many cases, a human fail to justify how? Unless the aim and objective are known it's difficult to justify how.

3. How: The answer to how lies in technology.

Earlier, some researchers built Intelligent Tutoring Systems (ITSs). Some of them are ACT Programming Tutor (Anderson *et al.*, 1990) based on ACT (Acceptance and Commitment Therapy) theory; an experiential system for learning probability named Stat Lady (Shute, 1995) was designed based on AI statistical relational techniques. Dialog-based Tutoring

Systems (DTSs) for Research Methods Tutors (RMT). Exploratory Learning Environments (ELEs) and an exploratory environment QUerying the Expert (QUE) (Metzler and Martincic, 1998) were created using Natural language processing, Bayesian network and Markov decision trees. The human-computer interface with collaboration, brain-computer interface, personalized/adaptive learning and real-time MOOC predictive modeling (Le *et al.*, 2018) attained by machine learning and deep learning.

3. Giant companies like Amazon, Google and Facebook (Holmes *et al.*, 2019) have financed to develop AIED products. Artificial Intelligence in Education (AIED) Society is an interdisciplinary community at the frontiers which encompasses computer science, education and psychology domains. Between AIED companies such as Knewton and Carnegie Learning, Global Learning XPrize is entitled to firm software which permits children to take control of their learning. AI has been developed as fundamental, persistent and inevitable, even if frequently hidden. Every so often email spam filtering is known as an advanced computer program, Cortona as a personal assistant, Netflix as a recommendation system and Duolingo as a language learning app. After studying the perspective, it is understandable that the magic lies in AI; it is useful to study every aspect of AI from basic concepts to understand the education applications thoroughly. Figure 10 elaborates the working of each AI technique and its analysis in education.

F. AI Techniques used in teaching-learning

i. Machine Learning:

Machine learning is neither a database problem nor a simple computer program; it is a part of AI. A system that can learn in a changing environment is called an intelligent system. It incorporates various learning styles as shown in Figure 11. Though the ML term was invented in 1959, it started to embellish in the 1990s.

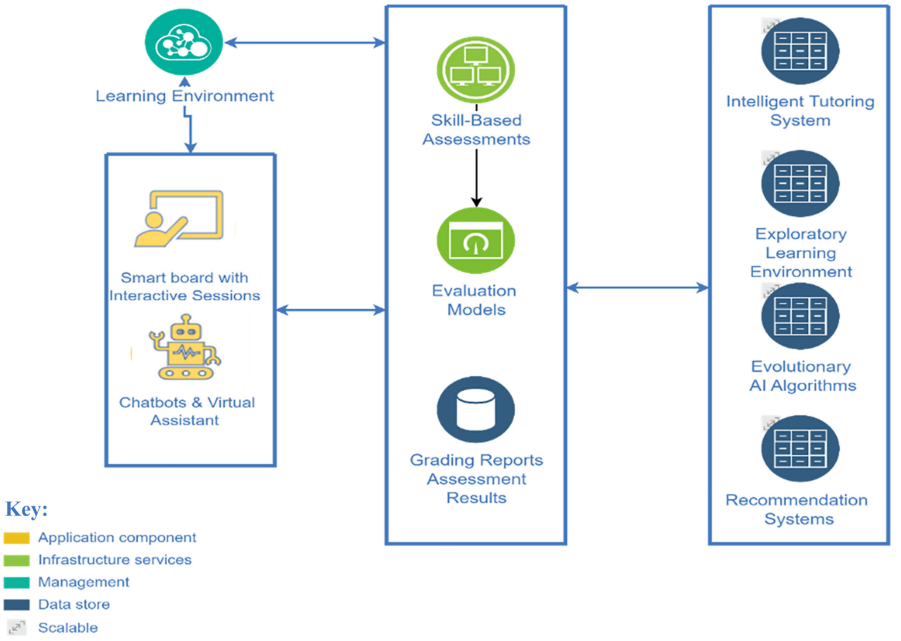


Figure 10.
AI cutting-edge education sector

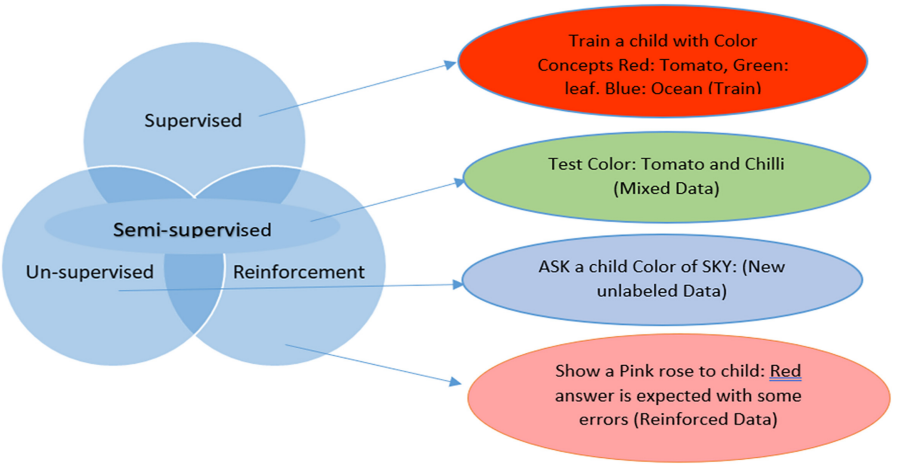


Figure 11.
Realm of Machine learning algorithms

In recent times, a few revisions of transformer-based architectures (Devlin *et al.*, 2018) based on Bidirectional Encoder Representations from Transformers (BERT) and neural network classifiers presented to realize short-answer grading assessment (Liu *et al.*, 2019; Sung *et al.*, 2017). Neural approaches have demonstrated across various domains and applications (NA) and masked language modeling (Camus and Filighera, 2020). If the dataset consists of approximately 1,000 training examples, we can adopt an embedded approach (classification with the neural network) to represent them using a non-neural classifier (Tassos and Natsis, 2011). AI-embedded computer programs, aid tutors and students. Explicitly educational innovation forces advancements in cognitive science and AI for learning (e.g. Crow *et al.*, 2018). There is a rational recommendation that validates the proficiency of numerous modern intelligent tutoring systems (Craig *et al.*, 2018). Current efforts are taken to build and use intelligent tutoring with collaborative learning (Olsen *et al.*, 2015); most intelligent tutoring systems focus on individual learning (student-centric education). For several years, research has shown gaming is harmful to students, especially teenagers, who spend more time with machines and less time with people (Oppenheimer, 2003). Online educational games have proven useful in some settings (González-González, 2014; Majdoub, 2016). AI as a routine has an artistic approach. On the other hand, getting completely rid of a human teacher from the process of student learning has risks. These are some of the educational strategies practiced in academics restricted to the purpose of education. Machine learning algorithms are listed in Table 6.

(1) Teacher-Centric Education (Teacher supportive)

(2) Small Group Education (Collaborative learning)

(3) Student-Centric Education (Student supportive)

(4) Active Learning

(5) Problem-Based Learning

(6) Project-Based Learning

(7) Inquiry-Based Learning

Parameter	Supervised (administered) learning (SL)	Unsupervised (unproven) learning (UL)	Semi-supervised (managed till some certain level) learning (SSL)	Reinforcement (supportive) learning (RL)
Advantages	1. To build accurate predictive models 2. To unite the bulk data to understand similarities and differences in a short time	1. Finding dissimilar groups or "clusters" within a data set is easy. 2. This requires fewer resources and faster computing	1. Visualize the interaction of variables and have a better understanding of the process 2. Basic probabilistic procedures are efficiently represented and implemented	1. As they learn, they can focus on the important parts of the space and ignore the rest; 2. Employ function approximation methods to simplify and learn faster
Education Model	Adaptive tutoring systems and assessments, Learning analytics (Chassignol *et al.*, 2018)	Collaborative learning, Project-based learning model, MOOC context (Sharma *et al.*, 2019)	Online educational games, chatbots and robotics (Pokrivcakova, 2019)	Automatic Curriculum Learning (Tassos and Natsis, 2011; Shuguang *et al.*, 2021)

Table 6. Usability of machine learning algorithms

(8) Cooperative Learning

(9) Flipped Classroom

4. The Knewton application makes real-time recommendations for students. It customizes course materials or content according to the learners' needs (Chassignol *et al.*, 2018). Chatbots (Mondal *et al.*, 2018) also improve the learning experiences of students because they leverage machine learning algorithms. Virtual Reality, 3-D technology in learning, includes usability, enjoyment, learner enthusiasm and motivation, to intensify interests in students (Tassos and Natsis, 2011, p. 773). Components of Adaptive and Intelligent Web-Based Educational Systems (AIWBES) are becoming popular among students (Srinivasan, 2022, p. 160). TurnItIn tools, (used for plagiarism check and revision assistance) and Pearson's Write-to-Learn tools help students in research and innovation (Halachev, 2013).

There exist some smart tutoring systems such as Carnegie Learning that rely on student's specific data to give personalized feedback to the student to directly work with.

ii. *Data Mining:* Knowledge discovery is the term used for data mining. The technique of applying machine learning methods to big data is called data mining.

- Data Mining: It is the process to build a suitable model or to develop an algorithm which is listed in Table 7 to extract knowledge by using the methods narrated in the above table. MOOT- Massive Open Online Text (MOOT) and Massive Open Online Courses (MOOC) use extensive data mining for knowledge interpretation. The example of whether a student gets university admission or not is portrayed in Figure 12.

iii. NLP is the accepted acronym for Neuro-Linguistic Programming (Bandler, 2020). It was the united idea of a mathematician and computer scientist, Richard Bandler and a professor in linguistics, John Grinder (Bandler and Grinder, 1975). Basic structure to process the language is depicted in Figure 13. Text Mining (Allahyari *et al.*, 2017) in

Sr. No.	Data mining method	Techniques	Purpose of application
1	Clustering	K-means, C-means, hierarchical clustering, decision trees, Markov models (Dutt *et al.*, 2015)	Slow, moderate and fast learners
2	Classification	Random forest, J48, Simple Cart, decision tree, Naïve Bays, SMO and Simple logistic (Dol, 2021)	Results and grade analysis
3	Outlier Detection, Regression	Linear, polynomial, logistic, ridge, Lasso regression (Srimani and Patil, 2014)	Predictive analysis: Failure and retention of students
4	Association Rule Mining	Apriori algorithm, FP (Frequent Pattern)-growth algorithm for the association rules mining (Ougiaroglou and Paschalis, 2012) multirelational association rule, generalized rule (Lei, 2022)	Self-assessment and collaborative Learning- think-pair-share
5	Pattern recognition, pattern matching	Feature selection, SVM, Matrix factorization, collaborative filtering (Zhang *et al.*, 2022)	Student performance-based system
6	Text mining/Concept mining/Web mining	Information retrieval, text summarization, ontologies (Ferreira-Mello *et al.*, 2019; Ahadi *et al.*, 2022)	Content analysis, educational research, automatically interact with students in forums

Table 7.
Strategic data mining techniques used in educational applications

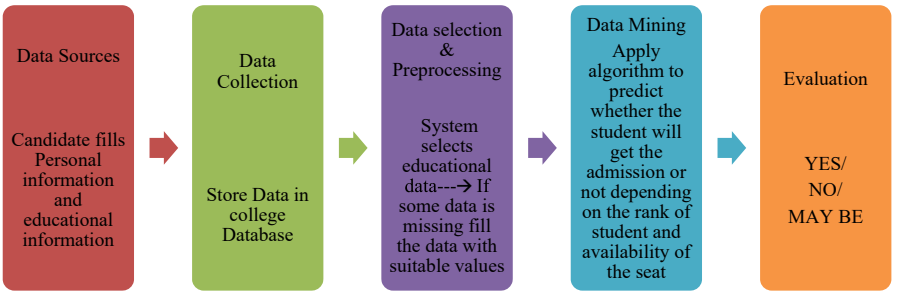


Figure 12.
Form filling process for admission

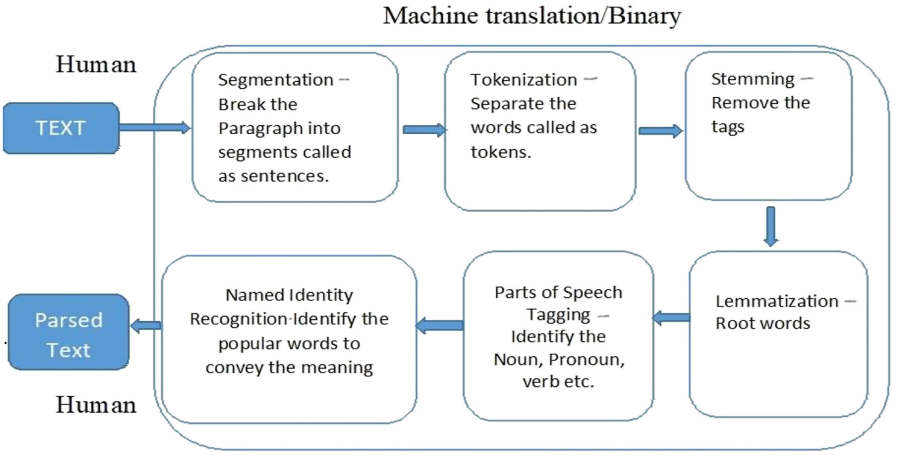


Figure 13.
Steps involved in natural language processing

Natural Language Processing (Sarker, 2022) is a process to extract meaningful information from various resources such as documents, articles, books, emails, reviews, blogs, comments, websites and podcasts.

Example: To study students' performance in response to the teaching, feedback on essays-OpenAI platform (Balaha and Saafan, 2021), question generation models (see Mostow *et al.*, n.d.), example generation of math problems, generating sentences that show different usages of the same word, showing examples of how text can be summarized, etc. (Chen *et al.*, 2020b; Srinivasan, 2022).

iv. Neural Networks (NN)

An excellent example of NN is DARPA Neural Network Study [DARP88] developed in 1988. It lists various neural network applications. It was launched through the adaptive channel equalizer in around 1984. This device is an owing commercial victory. It is the only neuron network used for long-distance telephone communication systems to stabilize voice signals (Martin *et al.*, 2014). The primary objective of studying NN here is to determine the appropriate connections to solve particular problems in an educational context.

v. *Genetics Algorithms:* [GA]- John Hollard developed the GA concept in the year 1956 inspired by Darwin's Theory. It portrays real biological evolution.

5. Genetic algorithms examples: Students result in a generation, predicting student academic performance, scheduling the timetables for the technical educational system, adaptive learning scheme generation for context-aware e-learning. The existing e-learning systems (Lee *et al.*, 2017) adapt content, based on various subsets of context parameters and generate customized learning paths.

The genetic algorithm is used for finding the optimal solution (Dang, 2022; Li and Song, 2022) which is based on a random probability distribution that may be analyzed statistically. GALA[GA3] educational system has been recognized to apply GA to dissimilar optimization problems. The system encodes a solution to the problem quickly and chooses the exact configuration of GA.

In existing conditions, these algorithms merge with AI, ML, Data mining, etc. Quite a few enhancements are needed to make GA more applicable in an educational context.

Figure 14 explains GA concept in a better and easy way. The foundation of GA is mathematical, which is illustrated with one of the methods, i.e. the Maximization method. The maximization method is demonstrated to understand the working principle of GA.

vi. Fuzzy Logic

Fuzzy set theory is invented by Zadeh (1965) to represent notions of the set in a better way. According to the Bezdek, computational intelligence depends on the numerical data provided by the manufacturers and not on the knowledge (Klement and Slany, 1993). Development of AI techniques plus cognitive science amplified the degree of "intelligence" in Computer-assisted Instruction (CAI) systems. They called as intelligent CAI (ICAI) and today these systems recognized as Intelligent Tutoring Systems.

There is the scope to modify the problems assessment model to work by means of varied areas or topics, besides field of linear programming.

1. Education should focus on value addition towards life; meaning quality education is needed. Fuzzy Logic is suitable for providing quality education (Alam and Pandey, 2017).

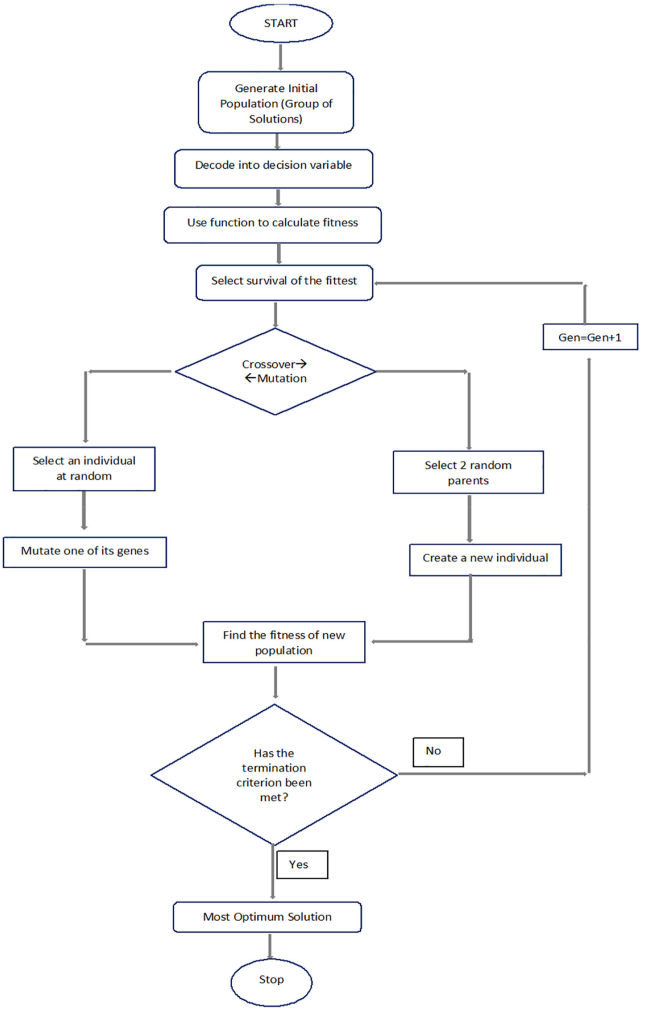


Figure 14.
Genetic algorithm flowchart

2. Computer Science and Engineering (CSE) characterizes the amalgamation of evolving problem resolving skills like fuzzy logic, genetic algorithms, statistical modeling, neural networks and evolutionary systems. Fuzzy logic-based model is capable to handle such uncertainties more vigorously as depicted in Figure 15.

Understanding the importance of the fuzzy logic, Garcia-Honrado (Salaki *et al.*, 2015), claims teaching of Fuzzy Logic in schools and high schools allows to see classical logic is a degenerative case of Fuzzy Logic.

By studying the application models, it is concluded that designated experimentations have a significant role in student performance. In the realm of 'AI in Education,' the integration of AI in assessment directly correlates with the cultivation of an enhanced learning styles, fostering a more resilient and flexible approach to acquiring and applying knowledge. It is supported by the literature demonstrating AI's ability to dynamically tailor assessments and evaluation strategies.

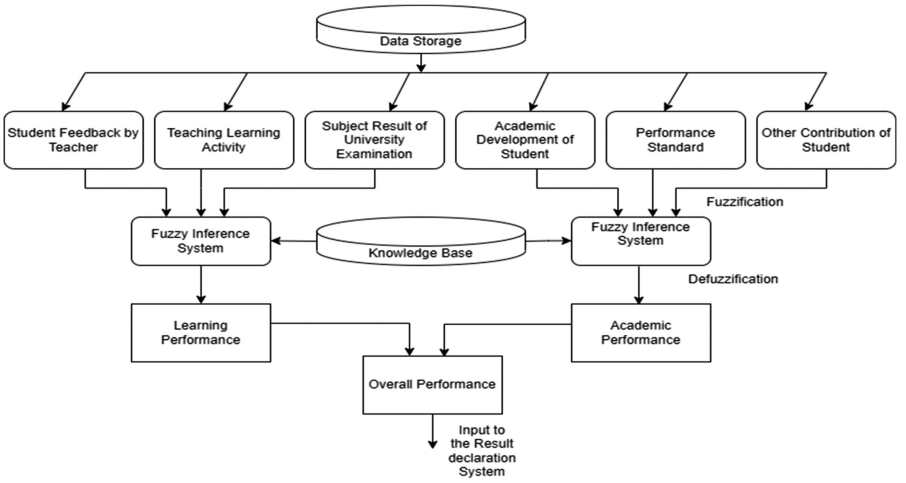


Figure 15.
Fuzzy logic framework

4. Conceptual framework: AI in HE assessment and its association with students' adversity quotient

Student is an inevitable component of education and assessment is a tool to measure student intelligence and intelligence has various dimensions. This section gives an overview of framework development steps.

A. AI-embedded education tools:

6. Education comprises of various models (Ouyang and Jiao, 2021), such as Domain Model, Pedagogy model, Learner model and Open learner model. Some tools are designed for specific purposes, for which users have to pay the fees or purchase. In Table 8 some of the popular AI-based educational tools are listed. These tools are not limited to the higher education but for different classes and age-groups.

B. Assessment: Assessment is an integral part of the education system, with a specific pattern as shown in Figure 16. Attention of the study depicts assessment types in Table 9. These types are in accordance with the program standards for undergraduate students which may vary depending on the specific outcomes.

Evaluation based research openings: After studying assessment evaluation and adversity as a skill, identified some of the research gaps as follows:

a. Education system is lacking in real-time adaptations in teaching and learning (Ferreira-Mello *et al.*, 2019)

b. Assessment and feedback practices can enhance students learning outcomes using AI (Lei, 2022).

Keeping these limitations in mind, the research focuses on bridging these gaps by relating AQ to the assessment patterns.

5. Adversity quotient

AQ is an individual's ability to overcome adversity and adapt to challenging situations (Matore *et al.*, 2021). It is a crucial skill for success in the modern world, where the ability to adapt to changing circumstances is essential. AQ (Wang *et al.*, 2022a, b) is defined by Stoltz (1997) as one's capability to fight with an approaching challenge, trouble, or problem, in

Table 8.
Educational tools

Sr. No.	Purpose	Tools/Technology	Domain	Features	Advantages	Limitations
1.	Grading, retention and performance prediction	1) WriteToLearn [Knowledge Analysis Technologies (KAT) engine] 2) Azure Cloud 3) Proctorio [Machine learning and facial detection technology]	Text Only Administration Proctor Base	Essay writing PowerBI Dashboards Cheating restriction	An online platform that focuses on enhancing writing abilities and improving reading comprehension	Not effective for developing higher-level writing skills, such as critical thinking and creativity
2.	Retention, dropout and performance prediction	1) CampusLabs [HTML5, jQuery and Google Analytics] 2) Nuro [Google Cloud infrastructure for AI] 3) ampusNexus Succeed [Microsoft Dynamics 365, machine learning and AI, Power BI.]	Guide Students Predictive analytics tracks student engagement	Pathways for Success Can be customized for institutes track progress	Provide institutions with data-driven insights into student engagement and performance	Issues with data accuracy and reliability
3.	Personalized Learning	1) Alta- Knewton [AI-driven adaptive learning technology] 2) Squirrel [AI] 3) MobyMax [G Suite, Skype and Zoom]	Math, Chinese, English, physics, all K-8 subjects	Personalized content, high quality after-school subjects, Widely adopted in the U. S	Adaptive learning platform to provide instructors with real-time data and analytics to track student progress	May not be suitable for all types of learners
4.	Sentiment Analysis	1) Pepper [Robotics, IoT] 2) NAO [programmable humanoid robot] 3) Talkwalker [AI]	Query-based, for special education, influencer identification	Effective conversation in administration, the students with disabilities, listening and analytic tool	Humanoid robots provide information and assistance in a user-friendly manner	Not be suitable for complex interactions or specialized knowledge, cost is relatively high

(*continued*)

Sr. No.	Purpose	Tools/Technology	Domain	Features	Advantages	Limitations
5.	Recommendation System	1) Qbot [world-first AI-infused technology] 2) MyEdMatch 3) TeacherMatch [AI for real-time analysis]	Content recommendation, talent recruitment, real-time analysis	Tags teachers and students to answer questions, help teachers in searching for jobs, assess teachers	Conversational agent reduces the workload of human customer service agents and improve customer satisfaction	Support limited by the quality and complexity of the underlying algorithms and natural language processing (NLP) models
6.	Classroom's Monitoring	1) Jibble [advanced AI face recognition technology] 2) Airclass [Augmented Reality] 3) SAFR (Secure Accurate Facial Recognition)	Attendance mechanism, school-specific, facial recognition	Tracks attendance with GPS, emotion recognition, allows entry in school after facial recognition	Technology is automated to identify and authenticate users quickly, without the need for manual review or verification	Still a risk of false positives (when the system incorrectly identifies a user) or false negatives, Cost
7.	Intelligent Tutoring system (Rus *et al.*, 2013)	1) DreamBox [AI, data analytics] 2) MATHia [AI and cognitive science] 3) Cognizant [evolutionary AI]	Math, Generic	Available in English and Spanish Mathematics specific, assessments and tests	Real-time feedback on students' progress	Focuses exclusively on math, Paid Platform

Table 8.

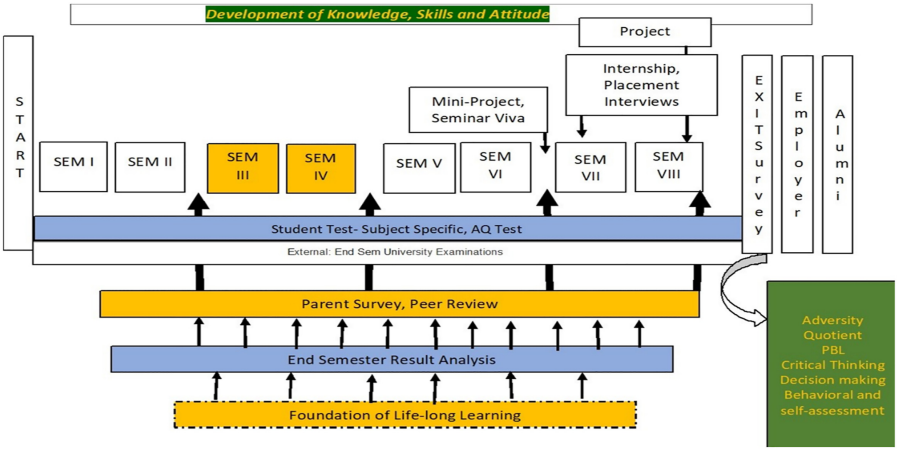


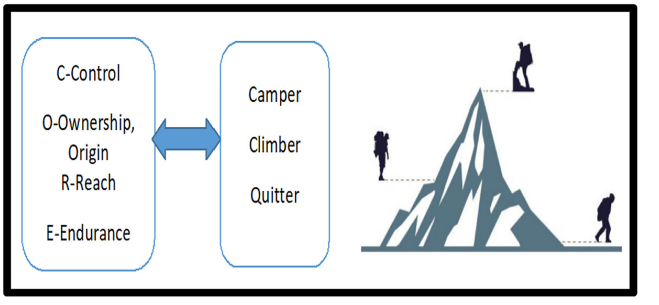
Figure 16.
Multimodal assessments conducted in higher education

Sr. No.	Assessment for learning: Ongoing and actionable	Assessment of learning: Grade-based	Assessment as learning: Actively involves students
Type of assessment	Formative Assessments	Summative Assessments	Ipsative Assessments
Methods used	1. Portfolios (work groups) 2. Group ventures 3. Improvement reports 4. Lecture discussions 5. Entry and exit tickets 6. Quick, regular quizzes	1. Exams 2. Portfolios 3. Final projects 4. Standardized tests	1. Intelligence Q. tests 2. On paper assessments 3. CET and GRE: Standardized college admissions tests

Table 9.
Various types of Assessment

addition turning it into a unique opportunity to get ahead successfully. In the current context of study, "AQ" is referred as an assessment of student's ability to meet problems, challenges, overcome existing challenges and turn them into an opportunity to accomplish success. Success refers to all-rounder achievement in academic training.

AQ is assessed in 3 categories based on scores as, less score = quitter, moderate score = climber and high score = camper as shown in Figure 17



Source(s): Stoltz (1997)

Figure 17.
Conceptual framework of AQ

To evaluate the academic performance, mapping of skill-based AQ assessment is the best tool. It is difficult to separate out AQ from IQ, EQ and SQ. Table 10 reveals the inter-relationship study to evaluate AQ with respect to other quotients. Concept of emotional intelligence (Salovey and Mayer, 1990), which includes the ability to perceive, understand and manage emotions. Zohar, D. and Marshall, I. discusses SQ as the ability to access higher meanings, values and purposes.

Metrics to affect adversity in education are:

1. Change of city: For higher education generally, students have to change city. They have to adapt new friends, place, food, environment etc. (Home sickness problem may occur because most of the students stay away from home to pursue higher education)
2. Medium of learning: Marathi to English, offline to online and online to hybrid, late time college schedules, attendance mandatory constraint, subjective to objective exams.

Case Study:

In Maharashtra (India) commission of competitive exams (MPSC) has decided to change the pattern of examination. i.e. from objective to subjective. MPSC candidates (aspiring students) protested against this decision and the commission had to withdraw its original decision (Pandey, 2013, Feb 23). Due to lack of resilience in students, the exam commission has to postpone the decision till 2025.

3. Socioeconomic status (SES) (Pong and Lam, 2023): This is a measure of a family's or individual's economic and social position in relation to others. It includes factors such as

	IQ	EQ	SQ	AQ
Reference	Gardner (1983), Kaur and Singh (2016)	Salovey and Mayer (1990), Mayer *et al.* (2008), Kaur and Singh (2016)	Zohar and Marshall (2000), Kaur and Singh (2016)	Stoltz (1997), Li *et al.* (2019)
Concept	cognitive intelligence	Emotional intelligence	Spiritual intelligence	Adversity intelligence
Tools	Wechsler Adult Intelligence Scale (WAIS) or the Stanford-Binet Intelligence Scale	Mayer-Salovey-Caruso Emotional Intelligence Test (MSCEIT), Emotional Quotient Inventory (EQ-i) and Trait Emotional Intelligence Questionnaire (TEIQue)	Spiritual Intelligence Self-Report Inventory (SISRI), the Spiritual Intelligence Scale (SIS) and the King Inventory of Spiritual Intelligence (KISI)	Adversity quotient profile, adversity response profile (ARP) and the resilience scale for adults (RSA)
Validity and Reliability	ranging from 75% to 90%	80–90%	70%–90%	70%–85%
Relationship with academic and professional success	Yes	Yes	Yes	Yes
Cultural and contextual factors	cognitive abilities, including verbal and nonverbal reasoning, problem-solving and memory	self-awareness, empathy and relationship management	higher meanings of life, values and purposes	individual's adversity quotient

Table 10.
AQ beyond IQ, EQ and SQ

income, education and occupation and race and ethnicity. These are measures of a student's racial and ethnic background. This can be important as certain groups may face systemic barriers that impact their education.

4. English Language Learners (ELL): This is a measure of a student's ability to speak and understand English. Students who are ELL may face additional challenges in accessing education.
5. Academic achievement: This is a measure of a student's performance (Barat *et al.*, 2018) academic subjects such as reading, writing and math. Low academic achievement can be an indicator of low adversity in education. School climate is a measure of the social and emotional environment in a school. A positive school climate can support student success, while a negative school climate can be a barrier to learning.

6. Discussion

RQ1. How AI benefits education domain?

AI plays a significant role (Lee *et al.*, 2017) in the creation of knowledge sources, knowledge material, online and virtual classes, assessment, administration, feedback personalized learning, adaptive systems used in education. Section of AI techniques discusses the literature, that Machine learning classified into four types (Matore and Khairani, 2016) and each type of learning is used depending on the application (Sharma *et al.*, 2019; Chassignol *et al.*, 2018; Shuguang *et al.*, 2021). We have studied popular techniques such as ML, DM (Namoun and Alshanqiti, 2021), NN , 2014; Halachev, 2013), NLP (A. Mondal *et al.*, 2018; Deng and Liu, 2018), GA (Wang, 2022; Julia *et al.*, 2020) and Fuzzy logic (Salaki *et al.*, 2015; Ilahi *et al.*, 2018). Some of the studies (Wong, 2018; Balaha and Saafan, 2021; Deng and Liu, 2018; Höhle and Klement, 2012) show, though every technique has its own advantages and disadvantages, they are interrelated or there is a thin boundary where the techniques use mixed methodology. ML is related with NN and deep learning (Martin *et al.*, 2014). Some of the studies (Srinivasan, 2022; Zhang *et al.*, 2022; Widodo *et al.*, 2022) focus on intelligent and smart systems (Sarker, 2022). Few of the studies (Luckin *et al.*, 2017; Zhu *et al.*, 2020; Sharma, 2022) focus on assessment-based techniques used in education. Above investigations proved, that there has been expansion in the AI techniques which can be used for specific educational purpose. At the beginning of the lockdown due to COVID-19 pandemic, rise in technology usage seen, i.e. for online coaching (through Zoom, Google Meet) and remote data storage (Google Drive, Classroom). Significantly less work has been done in AI with teaching-learning to construct full proof systems.

RQ2. What are the multimodal assessments conducted in higher education?

Literature work in section 3 shows multimodal assessment, for example, viva, practical-program based, written-Multiple Choice Questions (MCQ), project implementation, seminar presentation. We cannot assess each type by using same parameters and rubric. There are diverse rubrics we need to consider while assessing a student. There are some research gaps (Zhang *et al.*, 2022; Ferreira-Mello *et al.*, 2019; Pokrivcakova, 2019) identified with respect to knowledge-based assessments and techniques used to make the process efficient lexicons (Aktar, 2021). AI-based approaches are used for formative as well as summative assessment. The recent literature work on new methods in interactive learning environments (Luckin *et al.*, 2017), scientific argument writing (Zhu *et al.*, 2020), AI-based summative and formative assessment (Swiecki *et al.*, 2022; Gardner *et al.*, 2021) employed in higher education together

for better performance. In spite we are living in 21st century, the technology does not guarantee higher accuracies in assessment and academician take effort in manual validation of tests. The challenges are recognizing and utilizing appropriate AI technique for fair assessment.

RQ3. What are the different metrics to affect adversity in education?

1. CO_2RE model by Paul Stoltz is applicable for a professional, it is suggested that the parameters (Matore *et al.*, 2021; Biswas, 2018) required to succeed in individual's life e.g. student could be same. Control, ownership, origin, endurance, behavior, communication, adaptability (Menon, 2022), emotional intelligence, academic performance (Safi'i *et al.*, 2021; NA) parameters are used to evaluate an individual's overall AQ score, which can help them understand their strengths and weaknesses in dealing with adversity and identify areas where they can improve.

RQ4. What is the relevance of AQ in Indian (eastern) context and western context?

Some potential areas of research in this topic can include: Investigating the relationship between AQ and academic performance among Indian students.

1. Examining the role of AQ in the career development and success of Indian professionals can include studying the impact of AQ on job performance (Virgana *et al.*, 2022), career adaptability and entrepreneurship (Matore *et al.*, 2021; Lee *et al.*, 2017) among Indian workers. Figure 18 reveals AQ related research is negligible in India. It is important to investigate AQ relates to stress, anxiety, depression and other psychological outcomes (Pong and Lam, 2023) among Indian individuals. Investigating the cultural and contextual factors (Skrzypinska, 2021) that influence AQ in India supports cultural values, societal expectations, to respond to adversity and develop resilience.

Overall, evaluating AQ in the Indian context (Rahayu, 2021; Andrade, 2019; Tiwari, 2021) can be a meaningful and important research topic. It can contribute to understanding how

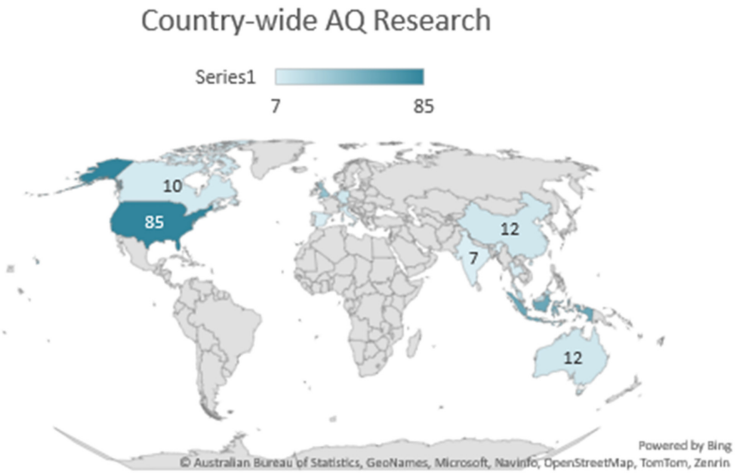


Figure 18.
AQ research in India

individuals in India respond to and cope with adversity and can have practical implications for improving education, mental health and career development outcomes.

RQ5. Mapping assessments with AQ would be a good step in higher education?

After discussing the above questions, literature revealed that there is a close relationship between academic performance (Safi'i *et al.*, 2021; and AQ. Without assessment (Bhatia, 2015; Hooda *et al.*, 2022)], academic performance cannot be evaluated. And academic performance refers not only to the academic grades. According to the NEP 2020, it is essential to assess not only the subject specific knowledge but also other factors affecting the student's achievement and success. Apart from the academics, one can embed self-assessment (Meijuan and Kaili, 2020; Yan *et al.*, 2020; Gardner, 1983), peer assessment (Naimnule *et al.*, 2020) and parent assessment to construct an AQ tool.

RQ6. Which are the various AI-based approaches to evaluate AQ?

In section 1, we reviewed various AI techniques contributing educational paradigm (Ougiaroglou and Paschalis, 2012). In section 4 and 5, study reveals assessments, AQ and CO_2RE model. There are some statistical methods such as Exploratory Factor Analysis, Confirmatory Factor Analysis and Rasch (Biswas, 2018) used for constructing AQ development tool. Development of intelligent machines to perform tasks which require human intelligence, corresponding to visual perception, speech recognition, decision-making and language translation give notion of AI. AQ is a measure of an individual's ability to cope with adversity and persevere through difficult situations.

AI and AQ are two different concepts that are not directly related. AI is scientific whereas AQ is psychological trait. Though AI and AQ are distinct concepts, there is potential for AI to be used to improve AQ and help individuals better cope with adversity. Overall study of literature suggests the following research gaps that should be addressed by our system.

Gap 1: Research collaboration Impact: Many researchers in psychology and education were unable to correlate AQ with IQ, EQ and SQ. Data should be generated by considering common factors which impact AQ.

Gap 2: Pedagogical connection impact: Lack of connections between teaching-learning analytics and AQ theories. There are limitations to 360-degree assessments in current higher education system.

Gap 3: "Competency based skills can't always be measured as zero or one" AI is the strong tool to assess natural intelligence of human brain.

7. Research areas to explore

Research on AQ can be categorized into three clusters.

a. *Relationship of AQ with IQ, EQ and SQ:* There is an established relationship between Adversity intelligence and other intelligences like cognitive, emotional and spiritual and explained in Table 10. The research focused on problem-solving skills, decision-making, critical thinking and adaptability i.e. ability to unlearn, accept the challenges and resilience.

b. Paul Stoltz Theory of AQ

AQ: Paul G. Stoltz has invented the AQ measurement depends on the CO2RE Model.

According to the Stoltz, AQ is not matter of black and white, yes or no, successful or unsuccessful. It is matter of degree. There are 60 items in Stoltz questionnaire some are with

+ sign and some items are with (minus) sign. The items (Stoltz, 1997) with minus sign should be considered to assess AQ.

Instruments: Adversity Response Profile (ARP) developed by Paul G. Stoltz. It has 2 potential weaknesses.

1. It has western culture context.
2. Designed for individuals in workplace environments (professionals).

c. AI methodology to evaluate AQ with the help of subject specific assessments and psychometric tests.

AQ Measurement for Students: The research can be conducted for any segment of education.

Table 11 trades of AQ assessment from general to specific target samples. Most of the AQ study focused on high school and college going students.

The concept of AQ is relatively new, and there is limited research on the use of AI approaches for evaluating AQ. However, there are several potential AI-based approaches that can be used to evaluate AQ.

8. Limitations

The education sector is highly diverse worldwide, whereas the focus of the present study is Indian higher education. The research specific literature investigation is extensively done from 2013 to 2023. Use of AI techniques in education and that is for evaluating AQ is an evolving area. It is very difficult to map multiple domains with single literature review, i.e. Education, AI, Assessments (Psychometric + Academics). Up-to-the-minute literary works were done in AI-based AQ evaluation for higher education students. Since it is an interdisciplinary emerging work, very less work is done in this area and therefore it is not possible to compare AI techniques based on AQ.

9. Conclusion and future work

The impact of AI techniques on teaching and learning has been the subject of analysis-oriented literature reviews, with sufficient sample sizes drawn from peer-reviewed journals and conference proceedings, resulting in statistics and conclusions on the use of AI in education. The papers reviewed related to AI for education open research opportunities in the field of psychometrics, educational assessment and AQ evaluation. This study highlights the potential for evaluating AQ for higher education students. However, research on AQ evaluation in the Indian context is still emerging, presenting a latent avenue for future

	Cluster 1	Cluster 2	Cluster 3
AQ instrument experimented on the various category which play a role in Education sector mostly	1997 to 2003 Principals at school level, university communal member group, leaders, students of junior high school	2004 to 2010 students of junior high school, high school undergraduate, vocational elementary school teachers, principals, drug addicts, traders, leaders and employees	2011 to the present blind schoolchildren, students of elementary, middle school, differently abled, principal of special needs students, autism spectrum (Ouyang and Jiao, 2021), leaders, school administrators, young mothers, nurses, special needs teachers

Table 11. Participant samples for AQ measure (Historical)

research. Investigating the relationship between AQ and academic performance among students is a crucial area of research. This can provide insights into the role of AQ in academic motivation, persistence and success in different academic disciplines and levels of education. AQ evaluation offers valuable insights into how individuals deal with and overcome challenges. The findings of this study have implications for higher education institutions to prepare for future challenges and better equip students with the necessary skills for success. Recent research suggests that there are numerous opportunities for using machine learning algorithms, such as sentiment analysis and transformer-based NLP approaches, to aid teachers in teaching literature, monitor student progress and assess personality measures.

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